

Outliers

(Univariate)

Definition → rare or extreme values (data points)

Detection

- Numerical
 - Z-score
 - IQR = $Q_3 - Q_1$
- Graphical

Treatment

- histogram
- pdf or cdf
- Boxplot

1.5 IQR

Q₁

Q₃

1.5 IQR

Multivariate

IQR

extend it to

Correlation matrix

Graphical

→ Heatmap

→ Scatter plots

→ Pair plots

Complex / high
dimension data

PCA

LOF
PYOD
AOD

Treatment

Remove →

Replace →

Imputation →

Range random

median

mean

percentile

Numeric data

Winsorization

Trimming

Use specific ML
algorithms